

7

Techniques of Integration



7.1

Integration by Parts

Integration by Parts

Every differentiation rule has a corresponding integration rule. For instance, the Substitution Rule for integration corresponds to the Chain Rule for differentiation. The rule that corresponds to the Product Rule for differentiation is called *integration by parts*.



Integration by Parts: Indefinite Integrals

Integration by Parts: Indefinite Integrals (1 of 3)

The Product Rule states that if f and g are differentiable functions, then

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

Integration by Parts: Indefinite Integrals (2 of 3)

In the notation for indefinite integrals this equation becomes

$$\int [f(x)g'(x) + g(x)f'(x)] dx = f(x)g(x)$$

or

$$\int f(x)g'(x) dx + \int g(x)f'(x) dx = f(x)g(x)$$

We can rearrange this equation as

$$1 \quad \int f(x)g'(x) dx = f(x)g(x) - \int g(x)f'(x) dx$$

Formula 1 is called the **formula for integration by parts**.

Integration by Parts: Indefinite Integrals (3 of 3)

It is perhaps easier to remember in the following notation.

Let $u = f(x)$ and $v = g(x)$. Then the differentials are $du = f'(x)dx$ and $dv = g'(x)dx$, so, by the Substitution Rule, the formula for integration by parts becomes

$$2 \quad \int u \, dv = uv - \int v \, du$$

Example 1

Find $\int x \sin x \, dx$.

Solution Using Formula 1:

Suppose we choose $f(x) = x$ and $g'(x) = \sin x$. Then $f'(x) = 1$ and $g(x) = -\cos x$.

(For g we can choose *any* antiderivative of g' .) Thus, using Formula 1, we have

$$\begin{aligned}\int x \sin x \, dx &= f(x)g(x) - \int g(x)f'(x) \, dx \\ &= x(-\cos x) - \int (-\cos x) \, dx \\ &= -x \cos x + \int \cos x \, dx \\ &= -x \cos x + \sin x + C\end{aligned}$$

Example 1 – Solution (1 of 2)

It's wise to check the answer by differentiating it. If we do so, we get $x \sin x$, as expected.

Solution Using Formula 2:

Let

$$u = x \quad dv = \sin x \, dx$$

Then

$$du = dx \quad v = -\cos x$$

Example 1 – Solution (2 of 2)

And so

$$\begin{aligned}\int x \sin x \, dx &= \int \overbrace{x}^u \overbrace{\sin x \, dx}^{dv} \\&= \overbrace{x}^u \overbrace{(-\cos x)}^v - \int \overbrace{(-\cos x)}^u \overbrace{dx}^{du} \\&= -x \cos x + \int \cos x \, dx \\&= -x \cos x + \sin x + C\end{aligned}$$



Integration by Parts: Definite Integrals

Integration by Parts: Definite Integrals (1 of 1)

If we combine the formula for integration by parts with Part 2 of the Fundamental Theorem of Calculus, we can evaluate definite integrals by parts. Evaluating both sides of Formula 1 between a and b , assuming f' and g' are continuous, and using the Fundamental Theorem, we obtain

$$6 \quad \int_a^b f(x)g'(x) dx = \left[f(x)g(x) \right]_a^b - \int_a^b g(x)f'(x) dx$$

Example 5

Calculate $\int_0^1 \tan^{-1}x \, dx$.

Solution:

Let

$$u = \tan^{-1}x \qquad dv = dx$$

Then

$$du = \frac{dx}{1+x^2} \qquad v = x$$

Example 5 – Solution (1 of 2)

So Formula 6 gives

$$\begin{aligned}\int_0^1 \tan^{-1} x \, dx &= x \tan^{-1} x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx \\&= 1 \cdot \tan^{-1} 1 - 0 \cdot \tan^{-1} 0 - \int_0^1 \frac{x}{1+x^2} \, dx \\&= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} \, dx\end{aligned}$$

Example 5 – Solution (2 of 2)

To evaluate this integral we use the substitution $t = 1 + x^2$ (since u has another meaning in this example). Then $dt = 2x \, dx$, so $dx = \frac{1}{2} dt$.

When $x = 0$, $t = 1$; when $x = 1$, $t = 2$; so

$$\begin{aligned}\int_0^1 \frac{x}{1+x^2} dx &= \frac{1}{2} \int_1^2 \frac{dt}{t} = \frac{1}{2} \ln |t| \Big|_1^2 \\ &= \frac{1}{2} (\ln 2 - \ln 1) = \frac{1}{2} \ln 2\end{aligned}$$

Therefore

$$\int_0^1 \tan^{-1} x \, dx = \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} dx = \frac{\pi}{4} - \frac{\ln 2}{2}$$



Reduction Formulas

Reduction Formulas (1 of 1)

The preceding examples show that integration by parts often allows us to express one integral in terms of a simpler one. If the integrand contains a power of a function, we can sometimes use integration by parts to reduce the power. In this way we can find a *reduction formula* as in the next example.

Example 6

Prove the reduction formula

$$7 \quad \int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

where $n \geq 2$ is an integer.

Solution:

Let

$$u = \sin^{n-1} x$$

$$dv = \sin x \, dx$$

Example 6 – Solution (1 of 2)

Then

$$du = (n - 1) \sin^{n-2} x \cos x \, dx \qquad v = -\cos x$$

and integration by parts gives

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n - 1) \int \sin^{n-2} x \cos^2 x \, dx$$

Since $\cos^2 x = 1 - \sin^2 x$, we have

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n - 1) \int \sin^{n-2} x \, dx - (n - 1) \int \sin^n x \, dx$$

Example 6 – Solution (2 of 2)

We solve this equation for the desired integral by taking the last term on the right side to the left side. Thus we have

$$n \int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n - 1) \int \sin^{n-2} x \, dx$$

or

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$



7.2

Trigonometric Integrals

Trigonometric Integrals

In this section we use trigonometric identities to integrate certain combinations of trigonometric functions.



Integrals of Powers of Sine and Cosine

Integrals of Powers of Sine and Cosine (1 of 4)

We begin by considering integrals in which the integrand is a power of sine, a power of cosine, or a product of these.

Example 2

Find $\int \sin^5 x \cos^2 x \, dx$.

Solution:

We could convert $\cos^2 x$ to $1 - \sin^2 x$, but we would be left with an expression in terms of $\sin x$ with no extra $\cos x$ factor.

Instead, we separate a single sine factor and rewrite the remaining $\sin^4 x$ factor in terms of $\cos x$:

$$\begin{aligned}\sin^5 x \cos^2 x &= (\sin^2 x)^2 \cos^2 x \sin x \\ &= (1 - \cos^2 x)^2 \cos^2 x \sin x\end{aligned}$$

Example 2 – Solution

Substituting $u = \cos x$, we have $du = -\sin x \, dx$ and so

$$\begin{aligned}\int \sin^5 x \cos^2 x \, dx &= \int (\sin^2 x)^2 \cos^2 x \sin x \, dx \\&= \int (1 - \cos^2 x)^2 \cos^2 x \sin x \, dx \\&= \int (1 - u^2)^2 u^2 (-du) = -\int (u^2 - 2u^4 + u^6) \, du \\&= -\left(\frac{u^3}{3} - 2 \frac{u^5}{5} + \frac{u^7}{7} \right) + C \\&= -\frac{1}{3} \cos^3 x + \frac{2}{5} \cos^5 x - \frac{1}{7} \cos^7 x + C\end{aligned}$$

Example 3

Evaluate $\int_0^{\pi} \sin^2 x \, dx$.

Solution:

If we write $\sin^2 x = 1 - \cos^2 x$, the integral is no simpler to evaluate. Using the half-angle formula for $\sin^2 x$, however, we have

$$\begin{aligned} \int_0^{\pi} \sin^2 x \, dx &= \frac{1}{2} \int_0^{\pi} (1 - \cos 2x) \, dx \\ &= \left[\frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) \right]_0^{\pi} \end{aligned}$$

Example 3 – Solution

$$\begin{aligned} &= \frac{1}{2} \left(\pi - \frac{1}{2} \sin 2\pi \right) - \frac{1}{2} \left(0 - \frac{1}{2} \sin 0 \right) \\ &= \frac{1}{2} \pi \end{aligned}$$

Notice that we mentally made the substitution $u = 2x$ when integrating $\cos 2x$.

Integrals of Powers of Sine and Cosine (2 of 4)

To summarize, we list guidelines to follow when evaluating integrals of the form $\int \sin^m x \cos^n x \, dx$, where $m \geq 0$ and $n \geq 0$ are integers.

Integrals of Powers of Sine and Cosine (3 of 4)

Strategy for Evaluating $\int \sin^m x \cos^n x \, dx$

(a) If the power of cosine is odd ($n = 2k + 1$), save one cosine factor and use $\cos^2 x = 1 - \sin^2 x$ to express the remaining factors in terms of sine:

$$\begin{aligned}\int \sin^m x \cos^{2k+1} x \, dx &= \int \sin^m x (\cos^2 x)^k \cos x \, dx \\ &= \int \sin^m x (1 - \sin^2 x)^k \cos x \, dx\end{aligned}$$

Then substitute $u = \sin x$.

(b) If the power of sine is odd ($m = 2k + 1$), save one sine factor and use $\sin^2 x = 1 - \cos^2 x$ to express the remaining factors in terms of cosine:

$$\begin{aligned}\int \sin^{2k+1} x \cos^n x \, dx &= \int (\sin^2 x)^k \cos^n x \sin x \, dx \\ &= \int (1 - \cos^2 x)^k \cos^n x \sin x \, dx\end{aligned}$$

Integrals of Powers of Sine and Cosine (4 of 4)

Then substitute $u = \cos x$. [Note that if the powers of both sine and cosine are odd, either (a) or (b) can be used.]

(c) If the powers of both sine and cosine are even, use the half-angle identities

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

It is sometimes helpful to use the identity

$$\sin x \cos x = \frac{1}{2} \sin 2x$$



Integrals of Powers of Secant and Tangent

Integrals of Powers of Secant and Tangent (1 of 6)

We can use a similar strategy to evaluate integrals of the form $\int \tan^m x \sec^n x \, dx$.

Since $\left(\frac{d}{dx}\right) \tan x = \sec^2 x$, we can separate a $\sec^2 x$ factor and convert the remaining (even) power of secant to an expression involving tangent using the identity $\sec^2 x = 1 + \tan^2 x$.

Or, since $\left(\frac{d}{dx}\right) \sec x = \sec x \tan x$, we can separate a $\sec x \tan x$ factor and convert the remaining (even) power of tangent to secant.

Example 5

Evaluate $\int \tan^6 x \sec^4 x \, dx$.

Solution:

If we separate one $\sec^2 x$ factor, we can express the remaining $\sec^2 x$ factor in terms of tangent using the identity $\sec^2 x = 1 + \tan^2 x$.

We can then evaluate the integral by substituting $u = \tan x$ so that $du = \sec^2 x \, dx$:

$$\int \tan^6 x \sec^4 x \, dx = \int \tan^6 x \sec^2 x \sec^2 x \, dx$$

Example 5 – Solution

$$\begin{aligned} &= \int \tan^6 x (1 + \tan^2 x) \sec^2 x \, dx \\ &= \int u^6 (1 + u^2) \, du = \int (u^6 + u^8) \, du \\ &= \frac{u^7}{7} + \frac{u^9}{9} + C \\ &= \frac{1}{7} \tan^7 x + \frac{1}{9} \tan^9 x + C \end{aligned}$$

Integrals of Powers of Secant and Tangent (2 of 6)

The preceding examples demonstrate strategies for evaluating integrals of the form $\int \tan^m x \sec^n x \, dx$ for two cases, which we summarize here.

Strategy for Evaluating $\int \tan^m x \sec^n x \, dx$

(a) If the power of secant is even ($n = 2k, k \geq 2$), save a factor of $\sec^2 x$ and use $\sec^2 x = 1 + \tan^2 x$ to express the remaining factors in terms of $\tan x$:

$$\begin{aligned} \int \tan^m x \sec^{2k} x \, dx &= \int \tan^m x (\sec^2 x)^{k-1} \sec^2 x \, dx \\ &= \int \tan^m x (1 + \tan^2 x)^{k-1} \sec^2 x \, dx \end{aligned}$$

Then substitute $u = \tan x$.

Integrals of Powers of Secant and Tangent (3 of 6)

(b) If the power of tangent is odd ($m = 2k + 1$), save a factor of $\sec x \tan x$ and use $\tan^2 x = \sec^2 x - 1$ to express the remaining factors in terms of $\sec x$:

$$\begin{aligned}\int \tan^{2k+1} x \sec^n x \, dx &= \int (\tan^2 x)^k \sec^{n-1} x \sec x \tan x \, dx \\ &= \int (\sec^2 x - 1)^k \sec^{n-1} x \sec x \tan x \, dx\end{aligned}$$

Then substitute $u = \sec x$.

Integrals of Powers of Secant and Tangent (4 of 6)

For other cases, the guidelines are not as clear-cut. We may need to use identities, integration by parts, and occasionally a little ingenuity.

We will sometimes need to be able to integrate $\tan x$ by using the formula given below:

$$\int \tan x \, dx = \ln |\sec x| + C$$

Integrals of Powers of Secant and Tangent (5 of 6)

We will also need the indefinite integral of secant:

$$1 \quad \int \sec x \, dx = \ln |\sec x + \tan x| + C$$

We could verify Formula 1 by differentiating the right side, or as follows. First we multiply numerator and denominator by $\sec x + \tan x$:

$$\int \sec x \, dx = \int \sec x \frac{\sec x + \tan x}{\sec x + \tan x} \, dx$$

Integrals of Powers of Secant and Tangent (6 of 6)

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$$

If we substitute $u = \sec x + \tan x$, then $du = (\sec x \tan x + \sec^2 x) dx$, so the integral becomes

$$\int \left(\frac{1}{u} \right) du = \ln |u| + C.$$

Thus we have

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

Example 7

Find $\int \tan^3 x \, dx$.

Solution:

Here only $\tan x$ occurs, so we use $\tan^2 x = \sec^2 x - 1$ to rewrite a $\tan^2 x$ factor in terms of $\sec^2 x$:

$$\begin{aligned}\int \tan^3 x \, dx &= \int \tan x \tan^2 x \, dx \\ &= \int \tan x (\sec^2 x - 1) \, dx \\ &= \int \tan x \sec^2 x \, dx - \int \tan x \, dx\end{aligned}$$

Example 7 – Solution

$$= \frac{1}{2} \tan^2 x - \ln |\sec x| + C$$

In the first integral we mentally substituted $u = \tan x$ so that

$$du = \sec^2 x \, dx.$$



Using Product Identities

Using Product Identities (1 of 1)

The following product identities are useful in evaluating certain trigonometric integrals.

2 To evaluate the integrals (a) $\int \sin mx \cos nx \, dx$, (b) $\int \sin mx \sin nx \, dx$, or (c) $\int \cos mx \cos nx \, dx$, use the corresponding identity:

$$(a) \sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$$

$$(b) \sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$(c) \cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

Example 9

Evaluate $\int \sin 4x \cos 5x \, dx$.

Solution:

This integral could be evaluated using integration by parts, but it's easier to use the identity in Equation 2(a) as follows:

$$\begin{aligned}\int \sin 4x \cos 5x \, dx &= \int \frac{1}{2} [\sin(-x) + \sin 9x] \, dx \\ &= \frac{1}{2} \int (-\sin x + \sin 9x) \, dx \\ &= \frac{1}{2} (\cos x - \frac{1}{9} \cos 9x) + C\end{aligned}$$



7.3

Trigonometric Substitution

Trigonometric Substitution (1 of 8)

In finding the area of a circle or an ellipse, an integral of the form $\int \sqrt{a^2 - x^2} dx$ arises, where $a > 0$.

If the integral were $\int x\sqrt{a^2 - x^2} dx$, the substitution $u = a^2 - x^2$ would be effective but, as it stands, $\int \sqrt{a^2 - x^2} dx$ is more difficult.

Trigonometric Substitution (2 of 8)

If we change the variable from x to θ by the substitution $x = a \sin \theta$, then the identity $1 - \sin^2 \theta = \cos^2 \theta$ allows us to get rid of the root sign because

$$\begin{aligned}\sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} \\ &= \sqrt{a^2 (1 - \sin^2 \theta)} \\ &= \sqrt{a^2 \cos^2 \theta} \\ &= a |\cos \theta|\end{aligned}$$

Trigonometric Substitution (3 of 8)

Notice the difference between the substitution $u = a^2 - x^2$ (in which the new variable is a function of the old one) and the substitution $x = a \sin \theta$ (the old variable is a function of the new one).

In general, we can make a substitution of the form $x = g(t)$ by using the Substitution Rule in reverse.

To make our calculations simpler, we assume that g has an inverse function; that is, g is one-to-one.

Trigonometric Substitution (4 of 8)

In this case, if we replace u by x and x by t in the Substitution Rule, we obtain

$$\int f(x) dx = \int f(g(t)) g'(t) dt$$

This kind of substitution is called *inverse substitution*.

We can make the inverse substitution $x = a \sin \theta$ provided that it defines a one-to-one function.

Trigonometric Substitution (5 of 8)

This can be accomplished by restricting θ to lie in the interval

$$\left[-\frac{\pi}{2}, \frac{\pi}{2} \right].$$

In the following table we list trigonometric substitutions that are effective for the given radical expressions because of the specified trigonometric identities.

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$

Table of Trigonometric Substitutions

Trigonometric Substitution (6 of 8)

In each case the restriction on θ is imposed to ensure that the function that defines the substitution is one-to-one.

Example 1

Evaluate $\int \frac{\sqrt{9-x^2}}{x^2} dx$.

Solution:

Let $x = 3 \sin \theta$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then $dx = 3 \cos \theta d\theta$ and

$$\begin{aligned}\sqrt{9-x^2} &= \sqrt{9-9\sin^2 \theta} \\ &= \sqrt{9\cos^2 \theta} \\ &= 3|\cos \theta| \\ &= 3\cos \theta\end{aligned}$$

(Note that $\cos \theta \geq 0$ because $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.)

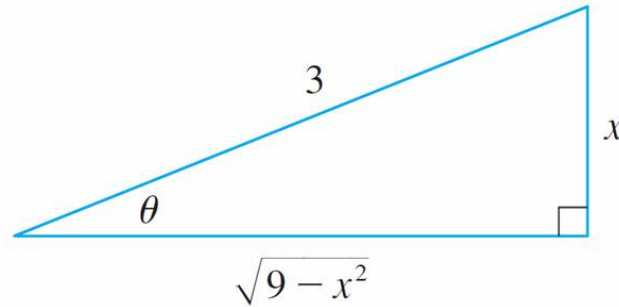
Example 1 – Solution (1 of 4)

Thus the Inverse Substitution Rule gives

$$\begin{aligned}\int \frac{\sqrt{9 - x^2}}{x^2} dx &= \int \frac{3 \cos \theta}{9 \sin^2 \theta} 3 \cos \theta d\theta \\&= \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta \\&= \int \cot^2 \theta d\theta \\&= \int (\csc^2 \theta - 1) d\theta \\&= -\cot \theta - \theta + C\end{aligned}$$

Example 1 – Solution (2 of 4)

Because this is an indefinite integral, we must return to the original variable x . This can be done either by using trigonometric identities to express $\cot \theta$ in terms of $\sin \theta = \frac{x}{3}$ or by drawing a diagram, as in Figure 1, where θ is interpreted as an angle of a right triangle.



$$\sin \theta = \frac{x}{3}$$

Figure 1

Example 1 – Solution (3 of 4)

Since $\sin \theta = \frac{x}{3}$, we label the opposite side and the hypotenuse as having lengths x and 3 .

Then the Pythagorean Theorem gives the length of the adjacent side as $\sqrt{9 - x^2}$, so we can simply read the value of $\cot \theta$ from the figure:

$$\cot \theta = \frac{\sqrt{9 - x^2}}{x}$$

(Although $\theta > 0$ in the diagram, this expression for $\cot \theta$ is valid even when $\theta < 0$.)

Example 1 – Solution (4 of 4)

Since $\sin \theta = \frac{x}{3}$, we have $\theta = \sin^{-1}\left(\frac{x}{3}\right)$ and so

$$\int \frac{\sqrt{9 - x^2}}{x^2} dx = -\cot \theta - \theta + C = -\frac{\sqrt{9 - x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C$$

Example 2

Find the area enclosed by the ellipse 橢圓

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Solution:

Solving the equation of the ellipse for y , we get

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} = \frac{a^2 - x^2}{a^2} \quad \text{or} \quad y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

Example 2 – Solution (1 of 5)

Because the ellipse is symmetric with respect to both axes, the total area A is four times the area in the first quadrant (see Figure 2).

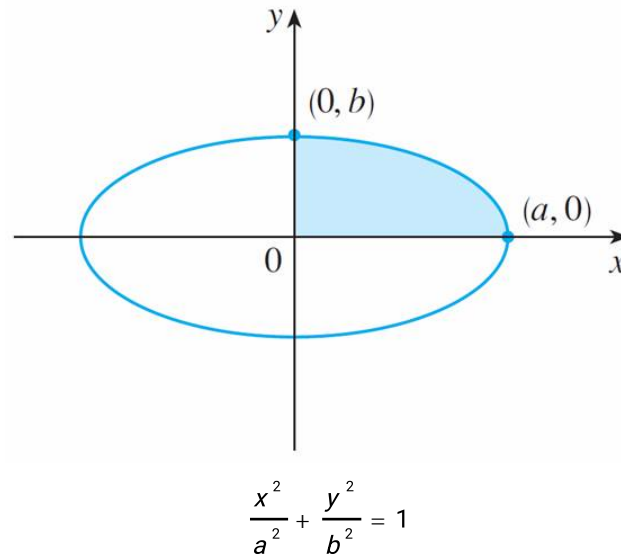


Figure 2

$$\begin{aligned}x &= a \sin \theta \\ \theta &\in [0, \frac{\pi}{2}] \\ dx &= a \cos \theta \, d\theta\end{aligned}$$

Example 2 – Solution (2 of 5)

The part of the ellipse in the first quadrant is given by the function

$$y = \frac{b}{a} \sqrt{a^2 - x^2} \quad 0 \leq x \leq a$$

and so

$$\frac{1}{4} A = \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx$$

To evaluate this integral we substitute $x = a \sin \theta$. Then $dx = a \cos \theta \, d\theta$.

Example 2 – Solution (3 of 5)

To change the limits of integration we note that when $x = 0$, $\sin \theta = 0$, so $\theta = 0$;

when $x = a$, $\sin \theta = 1$, so $\theta = \frac{\pi}{2}$.

Also

$$\begin{aligned}\sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} \\ &= \sqrt{a^2 \cos^2 \theta} \\ &= a |\cos \theta| \\ &= a \cos \theta\end{aligned}$$

since $0 \leq \theta \leq \frac{\pi}{2}$.

Example 2 – Solution (4 of 5)

Therefore

$$A = 4 \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx$$

$$= 4 \frac{b}{a} \int_0^{\pi/2} \underbrace{a \cos \theta} \cdot \underbrace{a \cos \theta \, d\theta}$$

$$= 4ab \int_0^{\pi/2} \cos^2 \theta \, d\theta \quad \text{even}$$

$$= 4ab \int_0^{\pi/2} \frac{1}{2} (1 + \cos 2\theta) \, d\theta$$

$$(\theta + \frac{1}{2} \sin 2\theta)' = 1 + \cos 2\theta$$

Example 2 – Solution (5 of 5)

$$\begin{aligned} &= 2ab\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{\pi/2} \\ &= 2ab\left(\frac{\pi}{2} + 0 - 0\right) \\ &= \pi ab \end{aligned}$$

We have shown that the area of an ellipse with semiaxes a and b is πab .

In particular, taking $a = b = r$, we have proved the famous formula that the area of a circle with radius r is πr^2 .

Trigonometric Substitution (7 of 8)

Note:

Because the integral in Example 2 was a definite integral, we changed the limits of integration and did not have to convert back to the original variable x .

Example 3

Find $\int \frac{1}{x^2 \sqrt{x^2 + 4}} dx$.

Solution:

Let $x = 2 \tan \theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then $dx = 2 \sec^2 \theta d\theta$ and

$$\begin{aligned}\sqrt{x^2 + 4} &= \sqrt{4(\tan^2 \theta + 1)} = \sqrt{4 \sec^2 \theta} \\ &= 2 |\sec \theta| \\ &= 2 \sec \theta\end{aligned}$$

Example 3 – Solution (1 of 3)

So we have

$$\begin{aligned}\int \frac{dx}{x^2 \sqrt{x^2 + 4}} &= \int \frac{2 \sec^2 \theta \, d\theta}{4 \tan^2 \theta \cdot 2 \sec \theta} \\ &= \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} \, d\theta\end{aligned}$$

To evaluate this trigonometric integral we put everything in terms of $\sin \theta$ and $\cos \theta$:

$$\frac{\sec \theta}{\tan^2 \theta} = \frac{1}{\cos \theta} \cdot \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{\cos \theta}{\sin^2 \theta}$$

Example 3 – Solution (2 of 3)

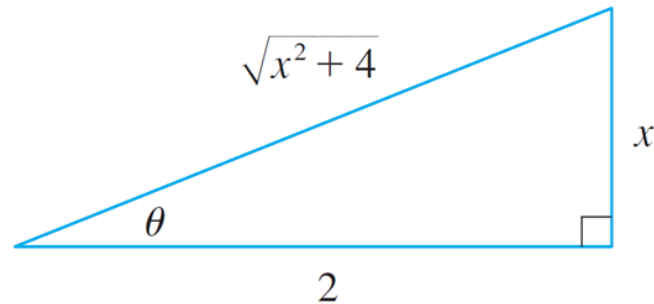
Therefore, making the substitution $u = \sin \theta$, we have

$$\begin{aligned}\int \frac{dx}{x^2 \sqrt{x^2 + 4}} &= \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta \\&= \frac{1}{4} \int \frac{du}{u^2} \\&= \frac{1}{4} \left(-\frac{1}{u} \right) + C \\&= -\frac{1}{4 \sin \theta} + C \\&= -\frac{\csc \theta}{4} + C\end{aligned}$$

Example 3 – Solution (3 of 3)

We use Figure 3 to determine that $\csc \theta = \frac{\sqrt{x^2 + 4}}{x}$ and so

$$\int \frac{dx}{x^2 \sqrt{x^2 + 4}} = -\frac{\sqrt{x^2 + 4}}{4x} + C$$



$$\tan \theta = \frac{x}{2}$$

Figure 3

Example 5

Evaluate $\int \frac{dx}{\sqrt{x^2 - a^2}}$, where $a > 0$.

Solution 1:

We let $x = a \sec \theta$, where $0 < \theta < \frac{\pi}{2}$ or $\pi < \theta < \frac{3\pi}{2}$.

Then $dx = a \sec \theta \tan \theta d\theta$ and

$$\begin{aligned}\sqrt{x^2 - a^2} &= \sqrt{a^2(\sec^2 \theta - 1)} \\ &= \sqrt{a^2 \tan^2 \theta} \\ &= a |\tan \theta| \\ &= a \tan \theta\end{aligned}$$

Example 5 – Solution 1 (1 of 3)

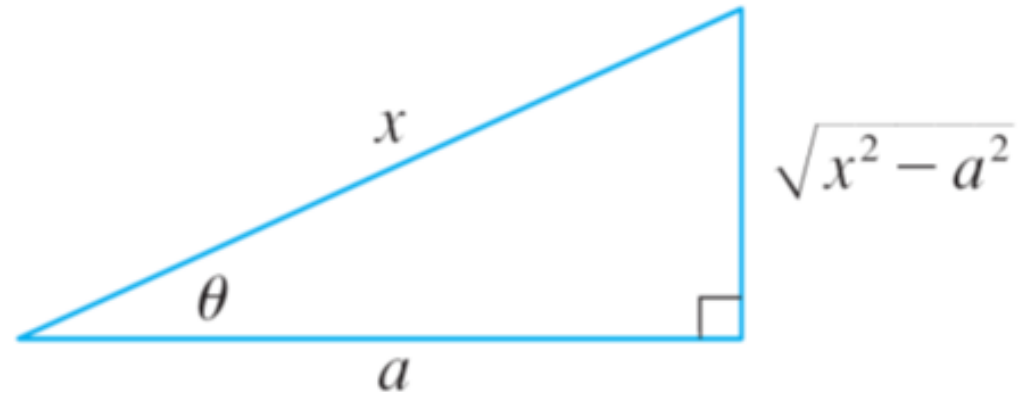
Therefore

$$\begin{aligned}\int \frac{dx}{\sqrt{x^2 - a^2}} &= \int \frac{a \sec \theta \tan \theta}{a \tan \theta} d\theta \\ &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C\end{aligned}$$

Example 5 – Solution 1 (2 of 3)

The triangle in Figure 4 gives $\tan \theta = \frac{\sqrt{x^2 - a^2}}{a}$, so we have

$$\begin{aligned}\int \frac{dx}{\sqrt{x^2 - a^2}} &= \ln \left| \frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right| + C \\ &= \ln \left| x + \sqrt{x^2 - a^2} \right| - \ln a + C\end{aligned}$$



$$\sec \theta = \frac{x}{a}$$

Figure 4

Example 5 – Solution 1 (3 of 3)

Writing $C_1 = C - \ln a$, we have

$$1 \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left| x + \sqrt{x^2 - a^2} \right| + C_1$$

Example 5 – Solution 2 (1 of 2)

For $x > 0$ the hyperbolic substitution $x = a \cosh t$ can also be used.

Using the identity $\cosh^2 y - \sinh^2 y = 1$, we have

$$\begin{aligned}\sqrt{x^2 - a^2} &= \sqrt{a^2(\cosh^2 t - 1)} \\ &= \sqrt{a^2 \sinh^2 t} \\ &= a \sinh t\end{aligned}$$

Example 5 – Solution 2 (2 of 2)

Since $dx = a \sinh t \, dt$, we obtain

$$\begin{aligned}\int \frac{dx}{\sqrt{x^2 - a^2}} &= \int \frac{a \sinh t \, dt}{a \sinh t} \\ &= \int dt \\ &= t + C\end{aligned}$$

Since $\cosh t = \frac{x}{a}$, we have $t = \cosh^{-1}\left(\frac{x}{a}\right)$ and

$$2 \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C$$

Trigonometric Substitution (8 of 8)

Note:

As Example 5 illustrates, hyperbolic substitutions can be used in place of trigonometric substitutions and sometimes they lead to simpler answers.

But we usually use trigonometric substitutions because trigonometric identities are more familiar than hyperbolic identities.

Example 6

Find

$$\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx.$$

Solution:

First we note that $(4x^2 + 9)^{3/2} = (\sqrt{4x^2 + 9})^3$ so trigonometric substitution is appropriate.

Although $\sqrt{4x^2 + 9}$ is not quite one of the expressions in the table of substitutions, it becomes one of them if we make the preliminary substitution $u = 2x$, which gives $\sqrt{u^2 + 9}$.

Example 6 – Solution (1 of 3)

Then we substitute $u = 3 \tan \theta$, or equivalently, $x = \frac{3}{2} \tan \theta$,

which gives $dx = \frac{3}{2} \sec^2 \theta \, d\theta$ and

$$\begin{aligned}\sqrt{4x^2 + 9} &= \sqrt{9 \tan^2 \theta + 9} \\ &= 3 \sec \theta\end{aligned}$$

When $x = 0$, $\tan \theta = 0$, so $\theta = 0$; when $x = \frac{3\sqrt{3}}{2}$, $\tan \theta = \sqrt{3}$, so $\theta = \frac{\pi}{3}$.

$$\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx = \int_0^{\pi/3} \frac{\frac{27}{8} \tan^3 \theta}{27 \sec^3 \theta} \frac{3}{2} \sec^2 \theta \, d\theta$$

Example 6 – Solution (2 of 3)

$$= \frac{3}{16} \int_0^{\pi/3} \frac{\tan^3 \theta}{\sec \theta} d\theta$$

$$= \frac{3}{16} \int_0^{\pi/3} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta$$

$$= \frac{3}{16} \int_0^{\pi/3} \frac{1 - \cos^2 \theta}{\cos^2 \theta} \sin \theta d\theta$$

Now we substitute $u = \cos \theta$ so that $du = -\sin \theta d\theta$. When $\theta = 0$, $u = 1$; when

$$\theta = \frac{\pi}{3}, \quad u = \frac{1}{2}.$$

Example 6 – Solution (3 of 3)

Therefore

$$\begin{aligned}\int_0^{\sqrt[3]{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx &= -\frac{3}{16} \int_1^{1/2} \frac{1 - u^2}{u^2} du \\&= \frac{3}{16} \int_1^{1/2} (1 - u^{-2}) du \\&= \frac{3}{16} \left[u + \frac{1}{u} \right]_1^{1/2} \\&= \frac{3}{16} \left[\left(\frac{1}{2} + 2 \right) - (1 + 1) \right] \\&= \frac{3}{32}\end{aligned}$$

7.4

Integration of Rational Functions by Partial Fractions

通过部分分式积分有理函数

Integration of Rational Functions by Partial Fractions (1 of 2)

In this section we show how to integrate any rational function (a ratio of polynomials) by expressing it as a sum of simpler fractions, called *partial fractions*, that we already know how to integrate.

To illustrate the method, observe that by taking the fractions $\frac{2}{(x-1)}$ and $\frac{1}{(x+2)}$ to a common denominator we obtain

$$\frac{2}{x-1} - \frac{1}{x+2} = \frac{2(x+2) - (x-1)}{(x-1)(x+2)} = \frac{x+5}{x^2+x-2}$$

Integration of Rational Functions by Partial Fractions (2 of 2)

If we now reverse the procedure, we see how to integrate the function on the right side of this equation:

$$\begin{aligned}\int \frac{x+5}{x^2+x-2} dx &= \int \left(\frac{2}{x-1} - \frac{1}{x+2} \right) dx \\ &= 2 \ln |x-1| - \ln |x+2| + C\end{aligned}$$

The Method of Partial Fractions

1. 分解有理函数 $\frac{P(x)}{Q(x)}$: $P(x), Q(x)$ 为多项式, 且 $\deg(P) < \deg(Q)$
2. 确定部分分式系数 : 由 $Q(x)$ 的根确定部分分式
3. 积分 : 对每个部分
 γ $Q(x)$ 有实根 a , $\frac{A}{x-a}$
 复根 $a \pm bi$, $\frac{Bx+C}{(x-a)^2+b^2}$

The Method of Partial Fractions (1 of 14)

To see how the method of partial fractions works in general, let's consider a **rational function**

有理函数(必须) $\rightarrow$ 两个多项式之比

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are **polynomials**. It's possible to express f as a sum of simpler fractions provided that the degree of P is less than the degree of Q . Such a rational function is called *proper*.

The Method of Partial Fractions (2 of 14)

We know that if

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \quad \text{多项式}$$

where $a_n \neq 0$, then the degree of P is n and we write $\deg(P) = n$.

If f is *improper*, that is, $\deg(P) \geq \deg(Q)$, then we must take the preliminary step of dividing Q into P (by long division) until a remainder $R(x)$ is obtained such that $\deg(R) < \deg(Q)$.

The Method of Partial Fractions (3 of 14)

The result is

$$1 \quad f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

where S and R are also polynomials.

As the next example illustrates, sometimes this preliminary step is all that is required.

真有理: 分母多项式阶次 > 分子多项式阶次

严格真有理: 严格大于

Example 1

Find $\int \frac{x^3 + x}{x - 1} dx$.

Solution:

Since the degree of the numerator is greater than the degree of the denominator, we first perform the long division.

This enables us to write

$$\begin{aligned}\int \frac{x^3 + x}{x - 1} dx &= \int \left(\underbrace{x^2 + x + 2} + \frac{2}{x - 1} \right) dx \\ &= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln |x - 1| + C\end{aligned}$$

$$\begin{array}{r}x^2 + x + 2 \\ x - 1 \overline{) x^3 + x} \\ \underline{x^3 - x^2} \\ x^2 + x \\ \underline{x^2 - x} \\ 2x + 0 \\ \underline{2x - 2} \\ 2\end{array}$$

The Method of Partial Fractions (4 of 14)

The next step is to factor the $Q(x)$ as far as possible.

It can be shown that any polynomial Q can be factored as a product of linear factors (of the form $ax + b$) and irreducible quadratic factors (of the form $ax^2 + bx + c$, where $b^2 - 4ac < 0$).

For instance, if $Q(x) = x^4 - 16$, we could factor it as

$$Q(x) = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$$

The Method of Partial Fractions (5 of 14)

The third step is to express the proper rational function $\frac{R(x)}{Q(x)}$ (from Equation 1) as a sum of **partial fractions** of the form

$$\frac{A}{(ax + b)^i} \quad \text{or} \quad \frac{Ax + B}{(ax^2 + bx + c)^j}$$

A theorem in algebra guarantees that it is always possible to do this. We explain the details for the four cases that occur.

The Method of Partial Fractions (6 of 14)

Case I The denominator $Q(x)$ is a product of distinct linear factors.

This means that we can write

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_kx + b_k)$$

where no factor is repeated (and no factor is a constant multiple of another).

The Method of Partial Fractions (7 of 14)

In this case the partial fraction theorem states that there exist constants $A_1, A_2, \dots, A_k$ such that

$$2 \quad \frac{R(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \dots + \frac{A_k}{a_kx + b_k}$$

These constants can be determined as in the following example.

Example 2

Evaluate $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$.

$$= x(2x^2 + 3x - 2) = x(2x - 1)(x + 2)$$

Solution:

Since the degree of the numerator is less than the degree of the denominator, we don't need to divide.

We factor the denominator as

$$\begin{aligned} 2x^3 + 3x^2 - 2x &= x(2x^2 + 3x - 2) \\ &= x(2x - 1)(x + 2) \end{aligned}$$

Example 2 – Solution (1 of 4)

Since the denominator has three distinct linear factors, the partial fraction decomposition of the integrand (2) has the form

$$3 \quad \frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

To determine the values of A , B , and C , we multiply both sides of this equation by the product of the denominators, $x(2x - 1)(x + 2)$, obtaining

$$4 \quad x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

Example 2 – Solution (2 of 4)

Expanding the right side of Equation 4 and writing it in the standard form for polynomials, we get

$$5 \quad x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x - 2A$$

The polynomials on each side of Equation 5 are identical, so their coefficients must be equal.

The coefficient of x^2 on the right side, $2A + B + 2C$, must equal the coefficient of x^2 on the left side—namely, 1.

Likewise, the coefficients of x are equal and the constant terms are equal.

Example 2 – Solution (3 of 4)

This gives the following system of equations for A , B , and C :

$$2A + B + 2C = 1$$

$$3A + 2B - C = 2$$

$$-2A = -1$$

Solving, we get, $A = \frac{1}{2}$, $B = \frac{1}{5}$, and $C = -\frac{1}{10}$, and so

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx = \int \left(\frac{1}{2} \frac{1}{x} + \frac{1}{5} \frac{1}{2x - 1} - \frac{1}{10} \frac{1}{x + 2} \right) dx$$

Example 2 – Solution (4 of 4)

$$= \frac{1}{2} \ln |x| + \frac{1}{10} \ln |2x - 1| - \frac{1}{10} \ln |x + 2| + K$$

In integrating the middle term we have made the mental substitution $u = 2x - 1$, which gives $du = 2dx$ and $dx = \frac{1}{2} du$.

The Method of Partial Fractions (8 of 14)

Note:

We can use an alternative method to find the coefficients A , B , and C in Example 2. Equation 4 is an identity; it is true for every value of x . Let's choose values of x that simplify the equation.

If we put $x = 0$ in Equation 4, then the second and third terms on the right side vanish and the equation then becomes $-2A = -1$, or $A = \frac{1}{2}$.

Likewise, $x = \frac{1}{2}$ gives $\frac{5B}{4} = \frac{1}{4}$ and $x = -2$ gives $10C = -1$, so $B = \frac{1}{5}$ and $C = -\frac{1}{10}$.

The Method of Partial Fractions (9 of 14)

Case II $Q(x)$ is a product of linear factors, some of which are repeated.

Suppose the first linear factor $(a_1x + b_1)$ is repeated r times; that is, $(a_1x + b_1)^r$ occurs in the factorization of $Q(x)$. Then instead of the single term

$$\frac{A_1}{(a_1x + b_1)}$$

in Equation 2, we would use

$$7 \quad \frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \cdots + \frac{A_r}{(a_1x + b_1)^r}$$

The Method of Partial Fractions (10 of 14)

By way of illustration, we could write

$$\frac{x^3 - x + 1}{x^2(x-1)^3} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} + \frac{D}{(x-1)^2} + \frac{E}{(x-1)^3}$$

but we prefer to work out in detail a simpler example.

Example 4

Find

$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx.$$

Solution:

The first step is to divide. The result of long division is

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{4x}{x^3 - x^2 - x + 1}$$

Example 4 – Solution (1 of 4)

The second step is to factor the denominator

$$Q(x) = x^3 - x^2 - x + 1.$$

Since $Q(1) = 0$, we know that $x - 1$ is a factor and we obtain

$$\begin{aligned}x^3 - x^2 - x + 1 &= (x - 1)(x^2 - 1) \\&= (x - 1)(x - 1)(x + 1) \\&= (x - 1)^2(x + 1)\end{aligned}$$

Example 4 – Solution (2 of 4)

Since the linear factor $x - 1$ occurs twice, the partial fraction decomposition is

$$\frac{4x}{(x-1)^2(x+1)} = \frac{\cancel{A}}{\cancel{x-1}} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

Multiplying by the least common denominator, $(x-1)^2(x+1)$, we get

$$8 \quad 4x = A(x-1)(x+1) + \underline{B(x+1) + C(x-1)^2}$$

Example 4 – Solution (3 of 4)

$$= (A + C)x^2 + (B - 2C)x + (-A + B + C)$$

Now we equate coefficients:

$$A + C = 0$$

$$B - 2C = 4$$

$$-A + B + C = 0$$

Example 4 – Solution (4 of 4)

Solving, we obtain $A = 1$, $B = 2$, and $C = -1$, so

$$\begin{aligned}\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx &= \int \left[x + 1 + \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1} \right] dx \\&= \frac{x^2}{2} + x + \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + K \\&= \frac{x^2}{2} + x - \frac{2}{x-1} + \ln \frac{x-1}{x+1} + K\end{aligned}$$

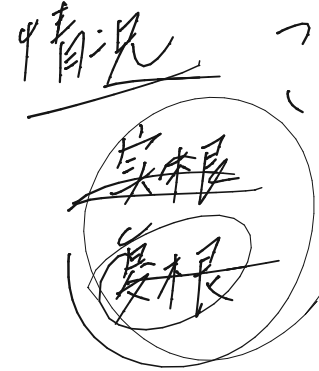
The Method of Partial Fractions (11 of 14)

Case III $Q(x)$ contains irreducible quadratic factors, none of which is repeated.

If $Q(x)$ has the factor $ax^2 + bx + c$, where $b^2 - 4ac < 0$, then, in addition to the partial fractions in Equations 2 and 7, the expression for $\frac{R(x)}{Q(x)}$ will have a term of the form

$$9 \quad \frac{Ax + B}{ax^2 + bx + c}$$

where A and B are constants to be determined.



The Method of Partial Fractions (12 of 14)

For instance, the function given by $f(x) = \frac{x}{[(x-2)(x^2+1)(x^2+4)]}$

has a partial fraction decomposition of the form

$$\frac{x}{(x-2)(x^2+1)(x^2+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{x^2+4}$$

The term given in (9) can be integrated by completing the square (if necessary) and using the formula

$$10 \quad \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

Example 6

Evaluate $\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx$.

Solution:

Since the degree of the numerator is *not less than* the degree of the denominator, we first divide and obtain

$$\frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} = 1 + \frac{x - 1}{4x^2 - 4x + 3}$$

Example 6 – Solution (1 of 3)

Notice that the quadratic $4x^2 - 4x + 3$ is irreducible because its discriminant $b^2 - 4ac = -32 < 0$. This means it can't be factored, so we don't need to use the partial fraction technique.

To integrate the given function we complete the square in the denominator:

$$4x^2 - 4x + 3 = (2x - 1)^2 + 2$$

This suggests that we make the substitution $u = 2x - 1$.

Example 6 – Solution (2 of 3)

Then $du = 2 dx$ and $x = \frac{1}{2}(u + 1)$, so

$$\begin{aligned}\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx &= \int \left(1 + \frac{x - 1}{4x^2 - 4x + 3} \right) dx \\ &= x + \frac{1}{2} \int \frac{\frac{1}{2}(u + 1) - 1}{u^2 + 2} du \\ &= x + \frac{1}{4} \int \frac{u - 1}{u^2 + 2} du\end{aligned}$$

Example 6 – Solution (3 of 3)

$$= x + \frac{1}{4} \int \frac{u}{u^2 + 2} du - \frac{1}{4} \int \frac{1}{u^2 + 2} du$$

$$= x + \frac{1}{8} \ln(u^2 + 2) - \frac{1}{4} \cdot \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{u}{\sqrt{2}}\right) + C$$

$$= x + \frac{1}{8} \ln(4x^2 - 4x + 3) - \frac{1}{4\sqrt{2}} \tan^{-1}\left(\frac{2x - 1}{\sqrt{2}}\right) + C$$

The Method of Partial Fractions (13 of 14)

Note:

Example 6 illustrates the general procedure for integrating a partial fraction of the form

$$\frac{Ax + B}{ax^2 + bx + c} \quad \text{where } b^2 - 4ac < 0$$

We complete the square in the denominator and then make a substitution that brings the integral into the form

$$\int \frac{Cu + D}{u^2 + a^2} du = C \int \frac{u}{u^2 + a^2} du + D \int \frac{1}{u^2 + a^2} du$$

Then the first integral is a logarithm and the second is expressed in terms of $\tan^{-1}$.

The Method of Partial Fractions (14 of 14)

Case IV $Q(x)$ contains a repeated irreducible quadratic factor.

If $Q(x)$ has the factor $(ax^2 + bx + c)^r$, where $b^2 - 4ac < 0$, then instead of the single partial fraction (9), the sum

$$(11) \quad \frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$$

occurs in the partial fraction decomposition of $\frac{R(x)}{Q(x)}$. Each of the terms in (11) can be integrated by using a substitution or by first completing the square if necessary.

Example 8

Evaluate $\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx$.

Solution:

The form of the partial fraction decomposition is

$$\frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}$$

Multiplying by $x(x^2 + 1)^2$, we have

$$-x^3 + 2x^2 - x + 1 = A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x$$

Example 8 – Solution (1 of 2)

$$= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex$$

$$= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A$$

If we equate coefficients, we get the system

$$A + B = 0 \quad C = -1 \quad 2A + B + D = 2 \quad C + E = -1 \quad A = 1$$

which has the solution $A = 1$, $B = -1$, $C = -1$, $D = 1$ and $E = 0$.

Example 8 – Solution (2 of 2)

Thus

$$\begin{aligned}\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} &= \int \left(\frac{1}{x} - \frac{x+1}{x^2+1} + \frac{x}{(x^2+1)^2} \right) dx \\&= \int \frac{dx}{x} - \int \frac{x}{x^2+1} dx - \int \frac{dx}{x^2+1} + \int \frac{x dx}{(x^2+1)^2} \\&= \ln|x| - \frac{1}{2} \ln(x^2+1) - \tan^{-1} x - \frac{1}{2(x^2+1)} + K\end{aligned}$$



Rationalizing Substitutions

Rationalizing Substitutions (1 of 1)

Some nonrational functions can be changed into rational functions by means of appropriate substitutions.

In particular, when an integrand contains an expression of the form $\sqrt[n]{g(x)}$, then the substitution $u = \sqrt[n]{g(x)}$ may be effective. Other instances appear in the exercises.

Example 9

Evaluate $\int \frac{\sqrt{x+4}}{x} dx$.

Solution:

Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = u^2 - 4$ and $dx = 2u du$. Therefore

$$\begin{aligned}\int \frac{\sqrt{x+4}}{x} dx &= \int \frac{u}{u^2 - 4} 2u du \\ &= 2 \int \frac{u^2}{u^2 - 4} du \\ &= 2 \int \left(1 + \frac{4}{u^2 - 4} \right) du\end{aligned}$$

Example 9 – Solution

We can evaluate this integral either by factoring $u^2 - 4$ as $(u - 2)(u + 2)$ and using partial fractions or by using Formula 6 with $a = 2$:

$$u^2 = x^2 + 4$$

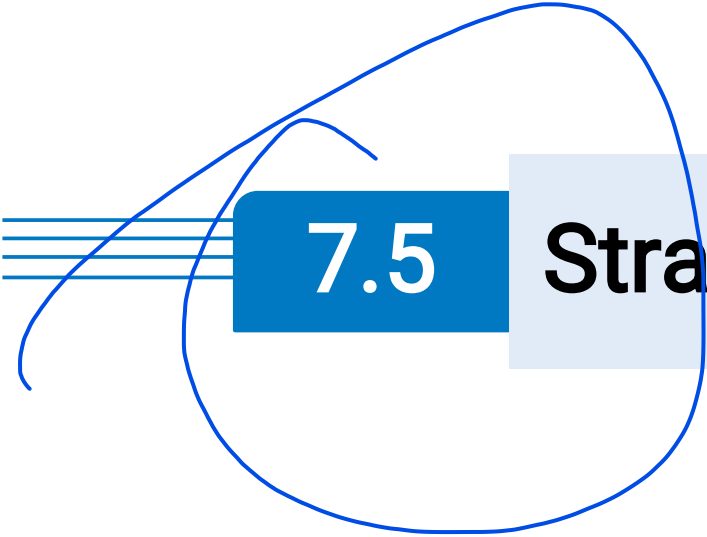
$$dx = 2u \, du$$

$$\int \frac{\sqrt{x+4}}{x} dx = 2 \int du + 8 \int \frac{du}{u^2 - 4} = (u+2)(u-2)$$

$$= 2u + 8 \cdot \frac{1}{2 \cdot 2} \ln \left| \frac{u-2}{u+2} \right| + C$$

$$= 2\sqrt{x+4} + 2 \ln \left| \frac{\sqrt{x+4}-2}{\sqrt{x+4}+2} \right| + C$$

$$\left(\frac{1}{4} \ln \left| \frac{u-2}{u+2} \right| \right)' = \frac{1}{u^2 - 4}$$

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7.5

Strategy for Integration



Guidelines for Integration

Guidelines for Integration (1 of 14)

In this section we present a collection of miscellaneous integrals in random order and the main challenge is to recognize which technique or formula to use.

No hard and fast rules can be given as to which method applies in a given situation, but we give some general guidelines that you may find useful.

A prerequisite for applying a strategy is a knowledge of the basic integration formulas.

In the table of Integration Formulas we have collected the integrals from our previous list together with several additional formulas that we have learned in this chapter.

Guidelines for Integration (2 of 14)

Table of Integration Formulas Constants of integration have been omitted.

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$$

$$2. \int \frac{1}{x} dx = \ln |x|$$

$$3. \int e^x dx = e^x$$

$$4. \int b^x dx = \frac{b^x}{\ln b}$$

$$5. \int \sin x dx = -\cos x$$

$$6. \int \cos x dx = \sin x$$

Guidelines for Integration (3 of 14)

7. $\int \sec^2 x \, dx = \tan x$

8. $\int \csc^2 x \, dx = -\cot x$

9. $\int \sec x \tan x \, dx = \sec x$

10. $\int \csc x \cot x \, dx = -\csc x$

11. $\int \sec x \, dx = \ln |\sec x + \tan x|$

12. $\int \csc x \, dx = \ln |\csc x - \cot x|$

13. $\int \tan x \, dx = \ln |\sec x|$

14. $\int \cot x \, dx = \ln |\sin x|$

$$\frac{1}{\sin x}$$

Guidelines for Integration (4 of 14)

$$15. \int \sinh x \, dx = \cosh x$$

$$16. \int \cosh x \, dx = \sinh x$$

$$17. \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$$

$$18. \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right), \quad a > 0$$

$$*19. \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right|$$

$$*20. \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left| x + \sqrt{x^2 \pm a^2} \right|$$

Guidelines for Integration (5 of 14)

Most of these formulas should be memorized. It is useful to know them all, but the ones marked with an asterisk need not be memorized since they are easily derived.

Formula 19 can be avoided by using partial fractions, and trigonometric substitutions can be used in place of Formula 20.

Once you are armed with these basic integration formulas, if you don't immediately see how to attack a given integral, you might try the following four-step strategy.

Guidelines for Integration (6 of 14)

1. Simplify the Integrand if Possible

Sometimes the use of algebraic manipulation or trigonometric identities will simplify the integrand and make the method of integration obvious. Here are some examples:

$$\int \sqrt{x}(1 + \sqrt{x}) \, dx = \int (\sqrt{x} + x) \, dx$$

$$\begin{aligned} \int \frac{\tan \theta}{\sec^2 \theta} \, d\theta &= \int \frac{\sin \theta}{\cos \theta} \cos^2 \theta \, d\theta \\ &= \int \sin \theta \cos \theta \, d\theta = \frac{1}{2} \int \sin 2\theta \, d\theta \end{aligned}$$

Guidelines for Integration (7 of 14)

$$\begin{aligned}\int (\sin x + \cos x)^2 dx &= \int (\sin^2 x + 2 \sin x \cos x + \cos^2 x) dx \\ &= \int (1 + 2 \sin x \cos x) dx\end{aligned}$$

2. Look for an Obvious Substitution

Try to find some function $u = g(x)$ in the integrand whose differential, $du = g'(x) dx$, also occurs, apart from a constant factor. For instance, in the integral

$$\int \frac{x}{x^2 - 1} dx$$

we notice that if $u = x^2 - 1$, then $du = 2x dx$.

Guidelines for Integration (8 of 14)

Therefore we use the substitution $u = x^2 - 1$ instead of the method of partial fractions.

3. Classify the Integrand According to Its Form

If Steps 1 and 2 have not led to the solution, then we take a look at the form of the integrand $f(x)$.

- (a) *Trigonometric functions.* If $f(x)$ is a product of powers of $\sin x$ and $\cos x$, of $\tan x$ and $\sec x$, or of $\cot x$ and $\csc x$, then we use the substitutions.
- (b) *Rational functions.* If f is a rational function, we use the procedure involving partial fractions.

Guidelines for Integration (9 of 14)

- (c) *Integration by parts.* If $f(x)$ is a product of a power of x (or a polynomial) and a transcendental function (such as a trigonometric, exponential, or logarithmic function), then we try integration by parts, choosing u and dv .
- (d) *Radicals.* Particular kinds of substitutions are recommended when certain radicals appear.
- (i) If $\sqrt{x^2 + a^2}$, $\sqrt{x^2 - a^2}$, or $\sqrt{a^2 - x^2}$ occurs, we use a trigonometric substitution.
- (ii) If $\sqrt[n]{ax + b}$ occurs, we use the rationalizing substitution $u = \sqrt[n]{ax + b}$.
- More generally, this sometimes works for $\sqrt[n]{g(x)}$.

Guidelines for Integration (10 of 14)

4. Try Again

If the first three steps have not produced the answer, remember that there are basically only two methods of integration: substitution and parts.

- (a) *Try substitution.* Even if no substitution is obvious (Step 2), some inspiration or ingenuity (or even desperation) may suggest an appropriate substitution.
- (b) *Try parts.* Although integration by parts is used most of the time on products of the form described in Step 3(c), it is sometimes effective on single functions.

Guidelines for Integration (11 of 14)

(c) *Manipulate the integrand.* Algebraic manipulations (perhaps rationalizing the denominator or using trigonometric identities) may be useful in transforming the integral into an easier form. These manipulations may be more substantial than in Step 1 and may involve some ingenuity. Here is an example:

$$\begin{aligned}\int \frac{dx}{1 - \cos x} &= \int \frac{1}{1 - \cos x} \cdot \frac{1 + \cos x}{1 + \cos x} dx = \int \frac{1 + \cos x}{1 - \cos^2 x} dx \\ &= \int \frac{1 + \cos x}{\sin^2 x} dx = \int \left(\csc^2 x + \frac{\cos x}{\sin^2 x} \right) dx\end{aligned}$$

Guidelines for Integration (12 of 14)

- (d) *Relate the problem to previous problems.* When you have built up some experience in integration, you may be able to use a method on a given integral that is similar to a method you have already used on a previous integral. Or you may even be able to express the given integral in terms of a previous one.

Guidelines for Integration (13 of 14)

For instance, $\int \tan^2 x \sec x \, dx$ is a challenging integral, but if we make use of the identity $\tan^2 x = \sec^2 x - 1$, we can write

$$\int \tan^2 x \sec x \, dx = \int \sec^3 x \, dx - \int \sec x \, dx$$

and if $\int \sec^3 x \, dx$ has previously been evaluated, then that calculation can be used in the present problem.

Guidelines for Integration (14 of 14)

- (e) *Use several methods.* Sometimes two or three methods are required to evaluate an integral. The evaluation could involve several successive substitutions of different types, or it might combine integration by parts with one or more substitutions.

Example 1 (1 of 2)

$$\int \frac{\tan^3 x}{\cos^3 x} dx$$

In Step 1 we rewrite the integral:

$$\int \frac{\tan^3 x}{\cos^3 x} dx = \int \tan^3 x \sec^3 x dx$$

The integral is now of the form $\int \tan^m x \sec^n x dx$ with m odd.

Alternatively, if in Step 1 we had written

$$\int \frac{\tan^3 x}{\cos^3 x} dx = \int \frac{\sin^3 x}{\cos^3 x} \frac{1}{\cos^3 x} dx$$

Example 1 (2 of 2)

$$= \int \frac{\sin^3 x}{\cos^6 x} dx$$

then we could have continued as follows with the substitution $u = \cos x$:

$$\begin{aligned} \int \frac{\sin^3 x}{\cos^6 x} dx &= \int \frac{1 - \cos^2 x}{\cos^6 x} \sin x dx \\ &= \int \frac{1 - u^2}{u^6} (-du) \\ &= \int \frac{u^2 - 1}{u^6} du \\ &= \int (u^{-4} - u^{-6}) du \end{aligned}$$



Can We Integrate All Continuous Functions?

Can We Integrate All Continuous Functions? (1 of 4)

The functions that we have been studied here are called **elementary functions**.

These are the polynomials, rational functions, power functions (x^n), exponential functions (b^x), logarithmic functions, trigonometric and inverse trigonometric functions, hyperbolic and inverse hyperbolic functions, and all functions that can be obtained from these by the five operations of addition, subtraction, multiplication, division, and composition.

Can We Integrate All Continuous Functions? (2 of 4)

For instance, the function

$$f(x) = \sqrt{\frac{x^2 - 1}{x^3 + 2x - 1}} + \ln(\cosh x) - x e^{\sin 2x}$$

is an elementary function.

If f is an elementary function, then f' is an elementary function but $\int f(x) dx$ need not be an elementary function.

Consider $f(x) = e^{x^2}$.

Can We Integrate All Continuous Functions? (3 of 4)

Since f is continuous, its integral exists, and if we define the function F by

$$F(x) = \int_0^x e^{t^2} dt$$

then we know from Part 1 of the Fundamental Theorem of Calculus that

$$F'(x) = e^{x^2}$$

Thus $f(x) = e^{x^2}$ has an antiderivative F , but it has been proved that F is not an elementary function.

This means that no matter how hard we try, we will never succeed in evaluating $\int e^{x^2} dx$ in terms of the functions we know.

Can We Integrate All Continuous Functions? (4 of 4)

The same can be said of the following integrals:

$$\int \frac{e^x}{x} dx$$

$$\int \sin(x^2) dx$$

$$\int \cos(e^x) dx$$

$$\int \sqrt{x^3 + 1} dx$$

$$\int \frac{1}{\ln x} dx$$

$$\int \frac{\sin x}{x} dx$$

In fact, the majority of elementary functions don't have elementary antiderivatives.

You may be assured, though, that the integrals in the following exercises are all elementary functions.